

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{K}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$\Upsilon_1 k^2$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{9}$	$-\frac{1}{9}\sqrt{2}\Upsilon_2 k^2$	0	0		
$\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	$-\frac{1}{9}\sqrt{2}\Upsilon_2 k^2$	$\frac{2\Upsilon_2 k^2}{9}$	0	0		
$\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	$\frac{\Upsilon_3 k^2}{2}$	$\frac{\Upsilon_3 k^2}{\sqrt{2}}$		
$\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	$\frac{\Upsilon_3 k^2}{\sqrt{2}}$	$\Upsilon_3 k^2$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	0	0	
			$\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$\frac{\Upsilon_4 k^2}{2}$	

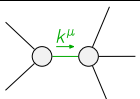
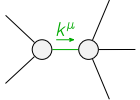
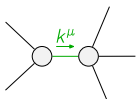
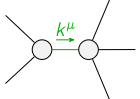
	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{J}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{J}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$\frac{1}{k^2 \binom{(4)}{\kappa_{15} \kappa_{16}}}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{1}{3 \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	$\frac{1}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$		
$\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	0	0	
			$\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^+)}$	

Abbreviations used in matrices

$$\begin{aligned} \Upsilon_1 &= \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} \text{ \&\& } \Upsilon_2 = 9 \binom{(4)}{\kappa_1} + 5 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \text{ \&\& } \Upsilon_3 = 2 \left(\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_{15}} \right) + \binom{(4)}{\kappa_{16}} \text{ \&\& } \\ \Upsilon_4 &= 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \text{ \&\& } \text{Det}(1^+) = \frac{1}{3} k^2 \left(9 \binom{(4)}{\kappa_1} + 5 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \right) \text{ \&\& } \text{Det}(2^+) = \frac{1}{2} k^2 \left(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} \right) \end{aligned}$$

Lagrangian

$$\begin{aligned} & \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^X \mathcal{K}^\alpha_{\alpha\beta} - \binom{(4)}{\kappa_1} \partial_X \mathcal{K}^\delta_{\beta\delta} \partial^X \mathcal{K}^\alpha_{\alpha\beta} - \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + \frac{8}{9} \binom{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} - \\ & \frac{5}{9} \binom{(4)}{\kappa_{16}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} - 4 \binom{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{40}{9} \binom{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{2}{9} \binom{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\ & \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \frac{10}{9} \binom{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \frac{14}{9} \binom{(4)}{\kappa_{16}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \\ & 6 \binom{(4)}{\kappa_1} \partial^X \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}^\delta_{\chi\beta} + 4 \binom{(4)}{\kappa_{15}} \partial^X \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}^\delta_{\chi\beta} + 2 \binom{(4)}{\kappa_{16}} \partial^X \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}^\delta_{\chi\beta} + 2 \binom{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\ & \frac{11}{9} \binom{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{10}{9} \binom{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \binom{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \binom{(4)}{\kappa_{16}} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1} = 0$	1	$\partial_\beta \mathcal{J}^\alpha{}^\beta = 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} = \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_X \partial^\alpha \mathcal{J}^\beta{}_\beta{}^X + \partial_X \partial^X \mathcal{J}^{\beta\alpha}{}_\beta = 0$	
$2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} = 0$	3	$\partial_X \mathcal{J}^{\alpha\beta\chi} + \partial_X \mathcal{J}^{X\alpha\beta} = \partial_X \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$	5	$3 \partial_\delta \partial_X \partial^\alpha \mathcal{J}^{X\beta\delta} + 3 \partial_\delta \partial_X \partial^\beta \mathcal{J}^{X\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^X{}^\delta{}_\chi = 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^X{}^\delta{}_\chi + 3 (\partial_\delta \partial^\delta \partial_X \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_X \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	12		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{81 \binom{(4)}{\kappa_1}^2 + 84 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 22 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (18 \binom{(4)}{\kappa_1}^2 + 19 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 5 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2)}$
	1	0	$-\frac{243 \binom{(4)}{\kappa_1}^2 + 180 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 34 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (18 \binom{(4)}{\kappa_1}^2 + 19 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 5 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2)}$
	1	0	$\frac{(180 \binom{(4)}{\kappa_1}^2 + 200 \binom{(4)}{\kappa_{15}}^2 + 200 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 50 \binom{(4)}{\kappa_{16}}^2 + 190 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) - \sqrt{3} \sqrt{(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 (4212 \binom{(4)}{\kappa_1}^2 + 4464 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 1207 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2)})}{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (18 \binom{(4)}{\kappa_1}^2 + 19 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 5 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2))}$
	1	0	$\frac{(180 \binom{(4)}{\kappa_1}^2 + 200 \binom{(4)}{\kappa_{15}}^2 + 200 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 50 \binom{(4)}{\kappa_{16}}^2 + 190 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + \sqrt{3} \sqrt{(2 \binom{(4)}{\kappa_1} + 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 (4212 \binom{(4)}{\kappa_1}^2 + 4464 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 1207 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2)})}{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (18 \binom{(4)}{\kappa_1}^2 + 19 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) + 5 (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2))}$
Resolved unitarity condition(s):	(Demonstrably impossible)		